Name: ____________________________

Date: ____________________________

Mr. Rawson • Y9 MYP Physics

1.3 – Period & Frequency of a Pendulum

Unit 1: Oscillations & Waves

Start here, and go slowly.
Criterion B is not a list of things a teacher marks at the end. Read it in order and it is **the investigation itself**: first you say what you are trying to find out, then what you think will happen and why, then what you will change and what you will keep the same, and only then how you will actually do it.

That order is not an accident. You cannot write a sensible method until you know what you are controlling, and you cannot know what to control until you have a question. Work down the criterion and it walks you through planning an experiment — which is why we do this part *before* we touch a pendulum.

Part 1 — Your investigation

Everything from here is **Criterion B — inquiring and designing**. Keep the **Criteria B and C** sheet open beside this one: each question below is labelled with the strand it is marked against.

Every group is testing something different. You all use the same pendulum and the same timing method. What changes is **the one thing you are allowed to change** — your **independent variable**. Write yours in the box below before you answer anything else.

Grp	Your variable	What you change, and how
1	**Length** of the string	Shortest to longest. Measure from the pivot to the *centre* of the bob, every time.
2	**Mass** of the bob	Same string length, same size of bob if you can — swap the mass only.
3	**Size** of the bob	Same mass, different diameter. Harder to arrange: ask before you start.
4	**Release angle** (displacement)	Same length, same bob. Release from a small angle, then wider. Use a protractor — "about the same" is not a measurement.
5	**Gravity**	**You cannot change gravity in this room.** Use the **PhET Pendulum Lab** simulation, which lets you set it — Earth, the Moon, Jupiter, and a slider in between.

Our group is number ______ **our independent variable is** ____________________________

Read this before you collect a single number.
"It made no difference" is a real result, and it is worth full marks. Some of the variables on that list genuinely change the period. Some genuinely do not. Nobody in the room knows which is which until the data is in — that is the entire point of testing.

A flat graph is a *finding*. It is not a failed experiment, and it is not a sign you did it wrong. **The one thing that will cost you marks is changing your numbers to make a line appear.**

Name: ______________________________ **Date:** ______________________

B.i What are we trying to find out?

1) **Describe** the problem you are testing. Write it as a question, with your own variable in it — then say, in a sentence, why anyone would want to know the answer. *B.i*

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Model answer: *Must contain THEIR variable and the period. "How does the mass of the bob affect the period of a pendulum?" Then a reason — clocks keep time on a pendulum, so knowing what does and does not change the period is what makes one reliable. At 7–8 the need is* described, *not just stated.*

B.ii What do you think will happen?

2) **Outline and explain** your hypothesis. Say what you think your variable will do to the period **and why** — a prediction with no reasoning behind it caps you at the bottom bands. *B.ii*

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Model answer: *Any honest prediction, WITH reasoning, is credited — it is tested, not marked right or wrong. "No change" is a perfectly good hypothesis and should be encouraged where the student can argue for it. Do not require $T = 2\pi\sqrt{L/g}$; they have not met it.*

Write it down now, and then leave it alone. A prediction you quietly edit once the results are in front of you is not a prediction — and being wrong costs you nothing here.

B.iii Variables, and the data you will collect

3) **State** your **independent** variable (the one you change) and your **dependent** variable (the one you measure). *B.iii*

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4) **Describe** how you will change it — *how many* different values, and spread how far apart? *B.iii*

Model answer: *At least five values, spread across a decent range, with repeats. "Sufficient, relevant data" is the phrase that separates 5–6 from 3–4 — three values close together is not sufficient.*

5) **State** your **control** variables — everything you must keep the same for the test to be fair. Include the things that are easy to forget. *B.iii*

Model answer: *Everything on the group list EXCEPT their own IV: length, bob mass, bob size, release angle. Plus the same counting method and the same person timing. Whichever variable they are testing is the one thing NOT controlled.*

6) **Describe** how you will measure the period — with what instrument, how many times, and how you will record it. *Write your best answer now; you will come back and improve it after Part 4.* *B.iii*

Model answer: *Time 20 swings on a stopwatch and divide by 20, repeated three times at each value, recorded in a ruled table with units in the headings. They have NOT practised this yet — the timing work is Part 4, deliberately after this. Accept a rough answer here and have them come back and sharpen it once Part 4 has shown them why one swing is not enough.*

B.iv Your method

7) **Design** your method. Number every step. List your equipment, and include one safety point. *B.iv*

Test it before you hand it in: give it to another group and see whether they could run it without asking you a single question. That is exactly what "logical, complete and safe" means on the criteria sheet.

Model answer: *Look for: equipment listed; numbered, ordered steps; the 20-swing rule stated; repeats; at least five values of their own variable; the controls actually held; one safety point (clamp stand weighted, swing path clear, feet clear).* ***Completeness*** *is what most students lose at 7–8, not safety.*

What happens next. You run the method you just wrote, and the numbers you get become **Criterion C — Part 2 of this sheet**. Your table, your graph, what it shows, and whether it agreed with the hypothesis you wrote before you started.

Then every group reports back. Between us we will have tested five different variables on the same pendulum, and **the pattern across all five groups is the actual physics** — more than any one group could find alone.

Part 2 — Your results

Everything from here is **Criterion C — processing and evaluating**. Same rule as before: each question is labelled with the strand it is marked against, so keep the **Criteria B and C** sheet beside you.

> **C is the half most people lose marks on, and it is the easier half.** B asks you to design something. C only asks you to **report honestly what happened** — but it has to be presented so that a stranger could read it without you standing there explaining.

C.i Presenting your data — the table

Here is what a results table should look like. This one is from the **spring experiment** you have already done.

Mass / g	Force / N	Extension trial 1 / cm	Extension trial 2 / cm	Extension trial 3 / cm	Mean extension / cm
0	0.00	0.0	0.0	0.0	0.0
100	0.98	3.7	4.1	3.9	3.9
200	1.96	8.0	7.8	8.1	8.0
300	2.94	11.5	11.7	11.6	11.6
400	3.92	15.9	15.7	16.0	15.9
500	4.90	19.4	19.6	19.5	19.5

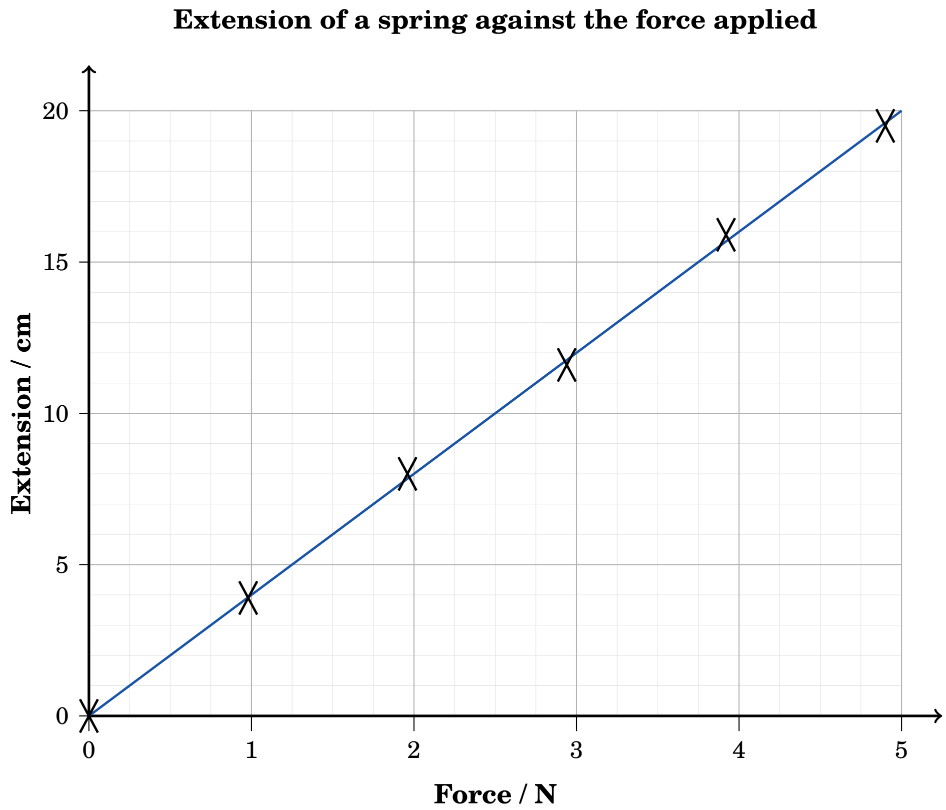
Why that table would score well.

- **Every heading has a unit**, written once in the heading — *not* beside each number. You never write "3.9 cm" in a cell if the heading already says **/ cm**.
- **Three trials**, then a **mean**. One reading is not data.
- **Every number to the same number of decimal places** down a column.
- **A zero row.** Nothing hanging on the spring, nothing stretched — it is a real reading and it belongs in the table.
- **The thing you changed is on the left**, what you measured is on the right.
- Ruled lines round every cell, so a reading cannot drift into the wrong column.

1) **Draw** your own results table below, for **your** variable. Head every column, put the unit in the heading, and leave room for three trials and a mean. *C.i*

C.i Presenting your data — the graph

And here is the same spring data, drawn as a graph.



Why that graph would score well.

- **It has a title** saying what is plotted against what.
- **Both axes are labelled, with units** — *Force / N*, *Extension / cm*.
- **The thing you changed is along the bottom**; what you measured goes up the side.
- **The scale uses most of the paper** and goes up in sensible steps. Squeezing the whole graph into one corner throws away accuracy.
- **Points marked with a small** $\times$, not fat dots you cannot read a value from.
- **One straight line of best fit, drawn with a ruler** — through the trend, with points either side of it. **Never dot-to-dot.**

2) **Draw** this **same** graph yourself, on a piece of graph paper, using the numbers from the table on the previous page. Then do the same for **your own** results. *C.i*

Six crosses and a ruled line. Every item on the green list above has to be on your version — that list is the mark scheme.

C.ii What does your data show?

3) Describe what your graph shows, in words. Say what happens to the period as your variable increases — and use physics to say it, not just "it goes up". *C.ii*

__Model answer:__ Must describe the SHAPE and say what it means. For length: as length increases the period increases, and the line curves rather than being straight. For mass, size or angle: the points lie on a flat line, so the period does not change — and that is the correct answer, credited fully. "Using scientific reasoning" at 5–6 means naming period, variable and the relationship, not just narrating the line.

4) State whether your points lie close to the line or scatter away from it, and what that tells you about your measurements. *C.ii*

__Model answer:__ Close to the line = consistent, repeatable timing. Wide scatter = timing error, usually reaction time on the stopwatch or counting swings wrongly.

C.iii Was your hypothesis right?

5) Discuss whether your data supports the hypothesis you wrote in Part 1. Quote your own numbers — do not just say "yes" or "no". *C.iii*

__Model answer:__ Must refer back to the actual prediction AND to real numbers from their table. "Based on the outcome of the investigation" is the phrase that separates 3–4 from 1–2 — a bare yes/no with no data quoted stays at the bottom. A disproved hypothesis loses NO marks; saying so clearly is what earns them.

C.iv Was your method sound?

6) **Discuss** how much you trust your results. Where could error have crept in, and how big was it compared with the differences you were measuring? *C.iv*

Model answer: *Expect: reaction time starting and stopping the stopwatch (about 0.2 s, which is why 20 swings are timed and divided); judging exactly when the bob passes the centre; measuring length to the centre of the bob rather than the top; releasing at a slightly different angle each time. Strong answers COMPARE the error with the effect — if the period changed by 0.05 s and the timing is good to 0.01 s, the effect is real.*

7) **State** one thing that was hard to keep the same, and what that could have done to your results. *C.iv*

C.v What would you change?

8) **Describe** two improvements or extensions, and say **why each one would make the investigation better** — an improvement with no reason attached does not score. *C.v*

Model answer: *Improvements: time more swings; repeat more times; use a photogate instead of a hand-held stopwatch; a denser bob to reduce air resistance; a protractor clamped in place for a repeatable release angle. Extensions: test a second variable, or push the range wider. At 7–8 each is DESCRIBED with a reason — "repeat more times, because it reduces the effect of one bad reading on the mean".*

> **Before you hand this in.** Read back through Part 2 with the criteria sheet open. For each of C.i to C.v, find the band you think you have reached and *say what would have moved you up one*. That question is the whole reason we started with the rubric instead of finishing with it.